# THE GRASSMANNIAN FUNCTOR G(K, N) OF QUOTIENTS IS REPRESENTABLE

THE STACKS PROJECT AUTHORS

*Editorial note.* Opening sheaf-property sentence carries 'some details omitted' (glueing of quotient data via Sheaves, Section glueing-sheaves); the four main steps are fully displayed.

*Editorial note.* Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 5221 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

*Allowed standard prerequisites (editorial).*

- Glue-functors criterion. schemes-lemma-glue-functors (cross-chapter, named). Provides the four-step representability template used for the whole proof.
- Right exactness of pullback. modules-lemma-exactness-pushforward-pullback (cross-chapter, named). Makes each chart subfunctor F_i stable under pullback.
- Surjectivity on stalks is open. modules-lemma-finite-type-surjective-on-stalk (cross-chapter, named). Opens up the chart locus U_i inside S.
- Nakayama. algebra-lemma-NAK (cross-chapter, named). Reduces chart membership to surjectivity on the residue fibre.

*Source reference.* Original reference 00DV: focused-seeds/geometry/S9-cons-gkn-representable/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original equation 089S: sources/stacks-constructions/pinned-original-source.tex, source lines 5180–5180. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 01AJ: focused-seeds/geometry/S9-cons-gkn-representable/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 01B8: focused-seeds/geometry/S9-cons-gkn-representable/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted

*Date*: stacks-project:constructions.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 089Q: focused-seeds/geometry/S9-cons-gkn-representable/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 01JH: focused-seeds/geometry/S9-cons-gkn-representable/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 01JJ: focused-seeds/geometry/S9-cons-gkn-representable/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 0098: focused-seeds/geometry/S9-cons-gkn-representable/ref-tags.txt, source lines 11–11. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 00AK: focused-seeds/geometry/S9-cons-gkn-representable/ref-tags.txt, source lines 12–12. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

**Lemma 0.1.** *Let $0 < k < n$. The functor $G(k, n)$ of (089S) is representable by a scheme.*

*Proof.* Set $F = G(k, n)$. To prove the lemma we will use the criterion of Schemes, Lemma 01JJ. The reason $F$ satisfies the sheaf property for the Zariski topology is that we can glue sheaves, see Sheaves, Section 00AK (some details omitted).

The family of subfunctors $F_i$. Let $I$ be the set of subsets of $\{1, \ldots, n\}$ of cardinality $n - k$. Given a scheme $S$ and $j \in \{1, \ldots, n\}$ we denote $e_j$ the global section

$$e_j = (0, \ldots, 0, 1, 0, \ldots, 0) \quad (1 \text{ in } j\text{th spot})$$

of $\mathcal{O}_S^{\oplus n}$. Of course these sections freely generate $\mathcal{O}_S^{\oplus n}$. Similarly, for $j \in \{1, \ldots, n-k\}$ we denote $f_j$ the global section of $\mathcal{O}_S^{\oplus n-k}$ which is zero in all summands except the $j$th where we put a 1. For $i \in I$ we let

$$s_i : \mathcal{O}_S^{\oplus n-k} \longrightarrow \mathcal{O}_S^{\oplus n}$$

which is the direct sum of the coprojections $\mathcal{O}_S \to \mathcal{O}_S^{\oplus n}$ corresponding to elements of $I$. More precisely, if $i = \{i_1, \ldots, i_{n-k}\}$ with $i_1 < i_2 < \ldots < i_{n-k}$ then $s_i$ maps $f_j$ to $e_{i_j}$ for $j \in \{1, \ldots, n-k\}$. With this notation we can set

$$F_i(S) = \{q : \mathcal{O}_S^{\oplus n} \to \mathcal{Q} \in F(S) \mid q \circ s_i \text{ is surjective}\} \subset F(S)$$

Given a morphism $f : T \to S$ of schemes the pullback $f^*s_i$ is the corresponding map over $T$. Since $f^*$ is right exact (Modules, Lemma 01AJ) we conclude that $F_i$ is a subfunctor of $F$.

Representability of $F_i$. To prove this we may assume (after renumbering) that $i = \{1, \ldots, n-k\}$. This means $s_i$ is the inclusion of the first $n-k$ summands. Observe that if $q \circ s_i$ is surjective, then $q \circ s_i$ is an isomorphism as a surjective map between finite locally free modules of the same rank (Modules, Lemma 089Q). Thus if $q : \mathcal{O}_S^{\oplus n} \to \mathcal{Q}$ is an element of $F_i(S)$, then we can use $q \circ s_i$ to identify $\mathcal{Q}$ with $\mathcal{O}_S^{\oplus n-k}$. After doing so we obtain

$$q : \mathcal{O}_S^{\oplus n} \longrightarrow \mathcal{O}_S^{\oplus n-k}$$

mapping $e_j$ to $f_j$ (notation as above) for $j = 1, \ldots, n-k$. To determine $q$ completely we have to fix the images $q(e_{n-k+1}), \ldots, q(e_n)$ in $\Gamma(S, \mathcal{O}_S^{\oplus n-k})$. It follows that $F_i$ is isomorphic to the functor

$$S \longmapsto \prod\nolimits_{j=n-k+1,\ldots,n} \Gamma(S, \mathcal{O}_S^{\oplus n-k})$$

This functor is isomorphic to the $k(n-k)$-fold self product of the functor $S \mapsto \Gamma(S, \mathcal{O}_S)$. By Schemes, Example 01JH the latter is representable by $\mathbf{A}^1_{\mathbf{Z}}$. It follows $F_i$ is representable by $\mathbf{A}^{k(n-k)}_{\mathbf{Z}}$ since fibred product over $\mathrm{Spec}(\mathbf{Z})$ is the product in the category of schemes.

The inclusion $F_i \subset F$ is representable by open immersions. Let $S$ be a scheme and let $q : \mathcal{O}_S^{\oplus n} \to \mathcal{Q}$ be an element of $F(S)$. By Modules, Lemma 01B8. the set $U_i = \{s \in S \mid (q \circ s_i)_s \text{ surjective}\}$ is open in $S$. Since $\mathcal{O}_{S,s}$ is a local ring and $\mathcal{Q}_s$ a finite $\mathcal{O}_{S,s}$-module by Nakayama's lemma (Algebra, Lemma 00DV) we have

$$s \in U_i \Leftrightarrow \big(\text{the map } \kappa(s)^{\oplus n-k} \to \mathcal{Q}_s/\mathfrak{m}_s\mathcal{Q}_s \text{ induced by } (q \circ s_i)_s \text{ is surjective}\big)$$

Let $f : T \to S$ be a morphism of schemes and let $t \in T$ be a point mapping to $s \in S$. We have $(f^*\mathcal{Q})_t = \mathcal{Q}_s \otimes_{\mathcal{O}_{S,s}} \mathcal{O}_{T,t}$ (Sheaves, Lemma 0098) and so on. Thus the map

$$\kappa(t)^{\oplus n-k} \to (f^*\mathcal{Q})_t/\mathfrak{m}_t(f^*\mathcal{Q})_t$$

induced by $(f^*q \circ f^*s_i)_t$ is the base change of the map $\kappa(s)^{\oplus n-k} \to \mathcal{Q}_s/\mathfrak{m}_s\mathcal{Q}_s$ above by the field extension $\kappa(t)/\kappa(s)$. It follows that $s \in U_i$ if and only if $t$ is in the corresponding open for $f^*q$. In particular $T \to S$ factors through $U_i$ if and only if $f^*q \in F_i(T)$ as desired.

The collection $F_i$, $i \in I$ covers $F$. Let $q : \mathcal{O}_S^{\oplus n} \to \mathcal{Q}$ be an element of $F(S)$. We have to show that for every point $s$ of $S$ there exists an $i \in I$ such that $s_i$ is surjective in a neighbourhood of $s$. Thus we have to show that one of the compositions

$$\kappa(s)^{\oplus n-k} \xrightarrow{s_i} \kappa(s)^{\oplus n} \to \mathcal{Q}_s/\mathfrak{m}_s\mathcal{Q}_s$$

is surjective (see previous paragraph). As $\mathcal{Q}_s/\mathfrak{m}_s\mathcal{Q}_s$ is a vector space of dimension $n-k$ this follows from the theory of vector spaces. □